CS340-070 Summer 2022
Algorithms and Data Structures

Time of Lectures	MTWR 9:30 - 10:45 am	Room: Classroom Bldg. 127
Instructor	Dr Ali Moltajaei Farid	Office: RIC 311 Email: afarid@uregina.ca
Office Hours	Friday 2:00 - 5:00 pm Monday 3:00 - 5:00 pm	
Tutorial	TBA	
Text	*Data Structures and Algorithms Analysis in C++, 4th edition, by Mark Allen Weiss*	
Useful Resources	*Learning Data Structures and Algorithms*	
Method of Evaluation	Assignments (4 x 5%)	20%
	Midterm Exam	30%
	Final Exam	50%

Topics

1. Review on Algorithm Analysis & Trees

2. Priority Queues

Binary heap, d-heap, Min-Max heap, leftist heap, binomial heap.

3. Sorting and Order Statistics

Overview and classification of sorting algorithms, insertion sort, mergesort, heapsort, quicksort, lower bound for sorting, bucket sort and radix sort, indirect sorting and external sorting, Order statistics.

4. Graphs and Networks

Implementation of graphs, topological sort, shortest path, network flow problems, minimum spanning tree, search techniques, introduction to NP-completeness.

5. Algorithm Design Techniques

Combinatorial optimization, greedy algorithms, divide and conquer, dynamic programming, randomized algorithms, backtracking algorithms, branch and bound.

6. Advanced Algorithms

Distributed Algorithms. Parallel Algorithms. Advanced algorithmic analysis: Amortized analysis, online and offline algorithms.

Policies

1. Information and class materials are available online through UR Courses. News and announcements will be posted on UR Courses, including UR Courses Email. It is the responsibility of students to regularly check their emails. Students are encouraged to turn on their UR Courses Email notifications.
2. URCourses should be used for all assignments submissions. It is the responsibility of the student to make sure that their submission has been successfully uploaded in UR Courses before the assignment due date. This can be checked by viewing the uploaded files. Email and Hardcopy submissions are not accepted.
3. Late assignments are not accepted for any reason and will receive 0 points, except for extensions granted to the entire class.
4. All your programming assignments must compile run with C on Hercules or Titan.
5. Any question regarding assignment submission or marking should be promptly addressed to the marker. In particular, any question/concern regarding assignment marking should be submitted to the marker no later than seven days after the marks are posted.
6. Advanced algorithmic and OO programming (in C++) skills are required for this course. These include: recursion, ADTs Lists, Stacks, Queues, and Trees.
7. You can discuss the assignment with other students but MAY NOT read, copy, or exchange other student's code. Students are encouraged to read the related Section of the Undergraduate Calendar on academic misconduct.
8. Attendance is mandatory in scheduled lectures hours. Little time is available to assist those who have missed these scheduled classes.

9. The final exam will be cumulative, but with more emphasis on material covered after the midterm.
10. **Please do not use URCourses to email the instructor; send an email directly with the above email address.**